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Introduction

Cattle Identification Using Muzzle Patterns

- This presentation explores an innovative approach to cattle identification, utilizing distinctive muzzle patterns.
- Muzzle patterns, akin to fingerprints, offer a unique and reliable means of distinguishing individual cattle.
- The little round-, oval-, or irregular-shaped protuberances on the nose area of cattle muzzle are defined as beads, while the elongated grooves and valleys arranged in a particular manner are defined as ridges . These uneven and distinct features make the muzzle identification for individual cattle possible.
- For identification a high-quality dataset having cattle muzzle images to evaluate and benchmark the performance of recognizing individual cattle with a variety of deep learning models is used. A total of 4923 muzzle images for 268 US feedlot finishing cattle (>12 images per animal on average) are taken with a mirrorless digital camera and processed to form the dataset

Significance and Potential Applications

- Precise Identification: Muzzle patterns provide a highly accurate and non-invasive way to differentiate cattle, enabling precision livestock management.
- Livestock Monitoring: The technology facilitates efficient monitoring of cattle populations, aiding in health assessment and behavior analysis.
- Breeding and Records: Muzzle pattern identification assists in maintaining accurate breeding records and pedigree tracking.
- Loss Prevention: In cases of theft or loss, this method proves invaluable for rapid cattle recovery.
- Agricultural Efficiency: Accurate identification enhances cattle-specific care and resource allocation, leading to optimized agricultural practices.

Literature review

AUTHOR / TITLE / YEAR	APPLIED METHODOLOGY / ALGORITHM USED	FINDINGS	RESULTS	LIMITATIONS
Biometric for cattle Identification using Muzzle Patterns/2020	Support vector machine (SVM)	Proposed method is evaluated using five different datasetsthree newly collected datasets of swamp buffalo captured in an outdoor field environment	Accuracy - 95%	Outdoor field environment includes challenges of freely moving cattle and differences in daylight
Shi-Tomasi corner detector for cattle identification from muzzle print image pattern / 2022	Bootstrapping Aggregation methodology along with Random Forest classifier and a combination of Shi-Tomasi, SURF, and SIFT features descriptor	Bootstrapping Aggregation methodology along with Random Forest classifier and a combination of Shi-Tomasi, SURF, and SIFT features descriptor.	Accuracy - 83.35%	The accuracy of the system is highly dependent on the quality of the input images. Poor lighting conditions, blurriness, or occlusions in the images can lead to reduced recognition accuracy.
Muzzle point pattern based techniques for individual cattle identification / 2017	LBP-Local Binary Patterns	The proposed automatic recognition algorithm based on muzzle point image patterns achieved a high identification accuracy of 93.87% with a database of 500 cattle, demonstrating the effectiveness of this approach for individual cattle identification	Accuracy - 93.87%	The high accuracy achieved in this study may be influenced by the quality and diversity of the database. The algorithm's performance in real-world, varied conditions and with larger databases may differ
Automatic Cattle Identification using YOLOv5 and Mosaic Augmentation: A Comparative Analysis / 2012	YOLOv5	YOLOv5 with transformer performed best with mean Average Precision (mAP) 0.5 (the average of AP when the IoU is greater than 50%) of 0.995, and mAP 0.5:0.95 (the average of AP from 50% to 95% IoU with an interval of 5%) of 0.9366.	Accuracy-80%	Couldn't identify exact patterns of beads and ridges

OPEN ISSUES

Data Quality
- Ensuring the dataset's accuracy and consistency.
- Addressing any potential noise or labeling errors.

Model Optimization
- Fine-tuning hyperparameters for better performance.
- Exploring alternative deep learning architectures.

Overfitting Mitigation
- Implementing robust techniques to prevent overfitting.
- Regularizing the model effectively.

Hardware Constraints
- Managing resource limitations during training.
- Considering scalable solutions for larger datasets.

Testing on New Data
- Extensively testing the model with diverse and challenging data.
- Ensuring the model's reliability in practical situations.

Scalability and Robustness
- Preparing for increasing cattle populations and diverse scenarios.
- Enhancing the model's robustness to various cattle conditions.

Deployment Challenges
- Ensuring smooth deployment in real-world applications.
- Addressing latency and scalability concerns

User Education
- Providing training and support to end-users of the cattle identification system.
- Ensuring they can make the most of the technology.

PROBLEM STATEMENT

Traditional cattle identification methods, including manual tagging and branding, pose several challenges in terms of efficiency, accuracy, and animal welfare. Manual tagging and branding are stressful for cattle and may have adverse effects on their well-being.These methods are labor-intensive, prone to errors, and often cause distress to the animals. In response to these issues, this model aims to develop an innovative solution: a non-invasive and automated cattle identification system based on the patterns of cattle muzzles.

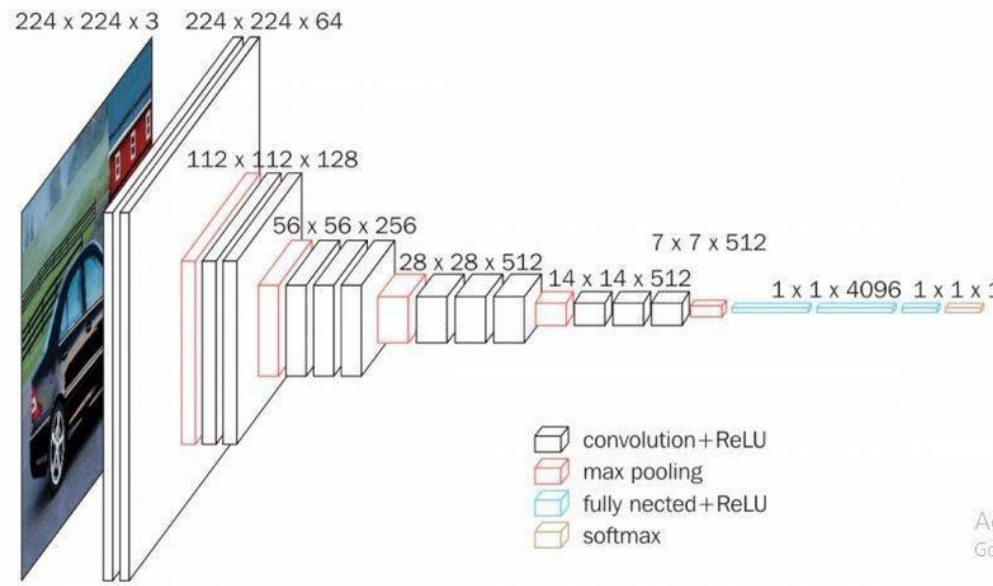
VGG16 Architecture

VGG16:

VGG16 refers to the VGG(visual geometry group) model, also called VGGNet. It is a convolution neural network (CNN) model supporting 16 layers. K. Simonyan and A. Zisserman from Oxford University proposed this model and published it in a paper called very deep convolution networks for largescale image recognition

The VGG16 model can achieve a test accuracy of 92.7% in ImageNet, a dataset containing more than 14 million training images across 1000 object classes. It is one of the top models from the ILSVRC-2014 competition.

VGG16 improves on AlexNet and replaces the large filters with sequences of smaller 3×3 filters.
VGG16, as its name suggests, is a 16-layer deep neural network.
VGG16 is thus a relatively extensive network with a total of 138 million parameters it’s huge even by today’s standards.
However, the simplicity of the VGGNet16 architecture is its main attraction.
The VGGNet architecture incorporates the most important convolution neural network features.



A VGG network consists of small convolution filters. VGG16 has three fully connected layers and 13 convolutional layers.

Here is a quick outline of the VGG architecture:

Input—VGGNet receives a 224×224 image input. In the ImageNet competition, the model's creators kept the image input size constant by cropping a 224×224 section from the center of each image.

Convolutional layers—the convolutional filters of VGG use the smallest possible receptive field of 3×3. VGG also uses a 1×1 convolution filter as the input's linear transformation.

ReLu activation—next is the Rectified Linear Unit Activation Function (ReLU) component, AlexNet's major innovation for reducing training time. ReLU is a linear function that provides a matching output for positive inputs and outputs zero for negative inputs.

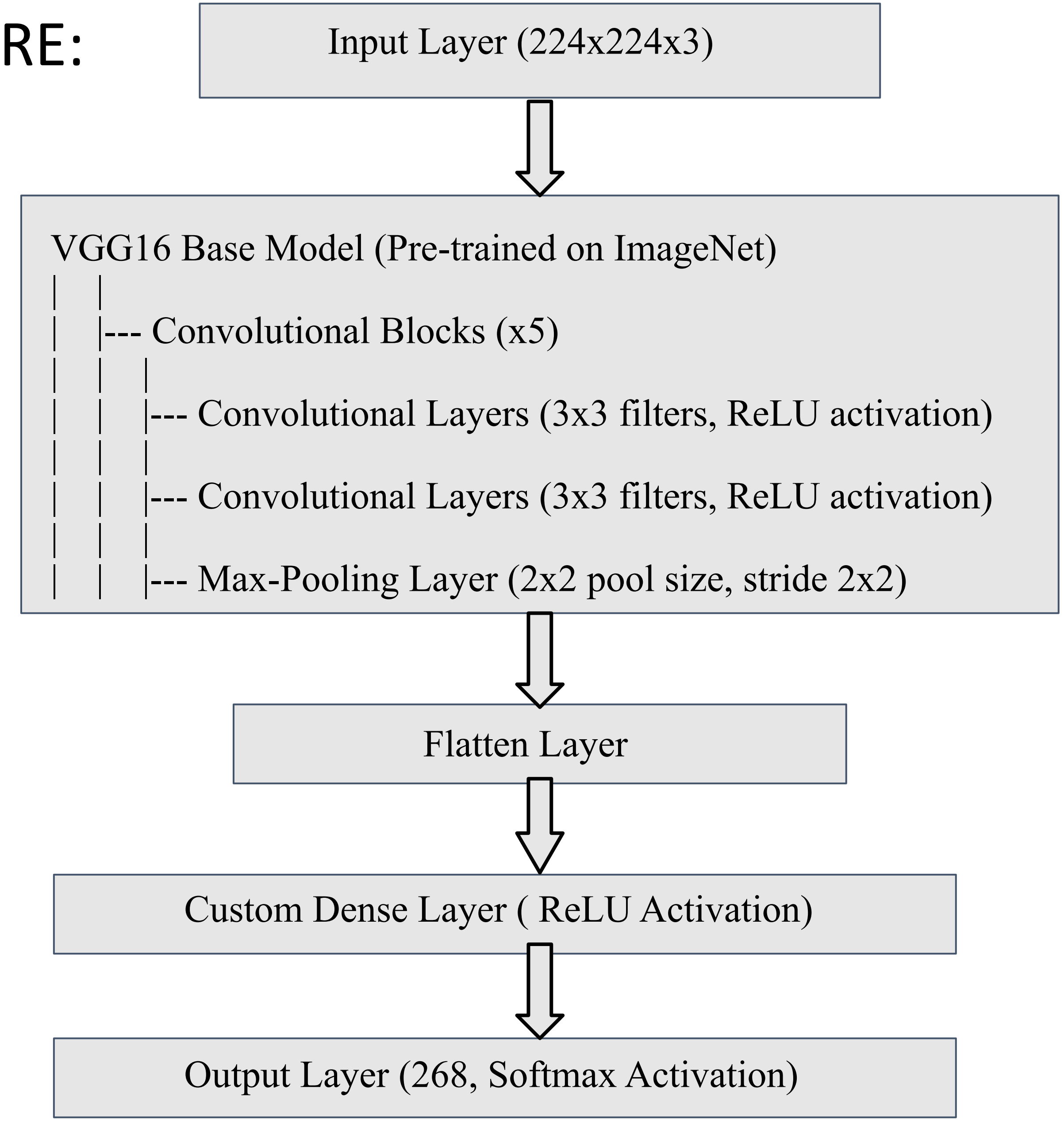
VGG has a set convolution stride of 1 pixel to preserve the spatial resolution after convolution (the stride value reflects how many pixels the filter "moves" to cover the entire space of the image).

Hidden layers—all the VGG network's hidden layers use ReLU instead of Local Response Normalization like AlexNet. The latter increases training time and memory consumption with little improvement to overall accuracy.

Pooling layers–A pooling layer follows several convolutional layers—this helps reduce the dimensionality and the number of parameters of the feature maps created by each convolution step. Pooling is crucial given the rapid growth of the number of available filters from 64 to 128, 256, and eventually 512 in the final layers.

Fully connected layers—VGGNet includes three fully connected layers. The first two layers each have 4096 channels, and the third layer has 1000 channels, one for every class.

PROPOSED ARCHITECTURE:



Functional & Non-Functional Requirements

Functional Requirements:

1. Image Input: The system should be able to accept images of cattle muzzles as input for identification.
2. Preprocessing: Implement image preprocessing techniques to enhance image quality and prepare it for feature extraction.
3. Feature Extraction: Use the VGG16 model to extract features from the input images.
4. Pattern Matching: Develop an algorithm that can match the extracted features to a database of known cattle muzzle patterns.
5. Real-time Processing: If required, specify that the system should process images in real-time or define acceptable processing time constraints..

Non-Functional Requirements:

1. Performance: Define the acceptable response time for identification requests. The system should respond within a specified time frame, considering the number of cattle records in the database.

2. Reliability: The system should be reliable, with minimal downtime or errors. Define a system availability target.

3. Accuracy: Specify the desired level of accuracy for cattle identification based on muzzle patterns. This could include a minimum accuracy rate or maximum acceptable error rate.

4. Scalability: Ensure that the system can handle a growing database of cattle records without a significant degradation in performance.

5. Usability: The user interface should be intuitive and user-friendly, making it easy for operators to input images and interpret identification results.

6. Maintainability: The system should be easy to maintain and update as needed, including updates to the VGG16 model or the database of cattle records.

7. Compliance: Ensure that the system complies with relevant laws and regulations regarding cattle identification and data privacy.

8. Resource Usage: Specify resource constraints, such as CPU and memory usage, to ensure the system can run efficiently on the target hardware.

Methodology

1. Data Collection: Gather a diverse dataset of cattle muzzle images, spanning different breeds and scenarios.

2. **Importing Libraries:** Necessary libraries are imported, including TensorFlow, which is used for building and training neural networks, and other modules for data handling and image preprocessing.

3. Data Preprocessing Steps

- Resizing: Standardize all images to a common size to ensure consistent input dimensions for the model.
- Normalization: Scale pixel values to a common range (e.g., [0, 1]) to facilitate convergence during training.

4. Data Augmentation for Enhanced Generalization

- Shearing, Zooming, and Flipping: Apply transformations to training images, expanding the dataset and increasing the model's robustness to variations.
- Validation Split: Divide the dataset into training and validation sets to assess the model's performance on unseen data.

5. Building the VGG16 Model and compiling it : Utilize transfer learning with the VGG16 model to exploit pre-trained features. Then the model is compiled taking loss-categorical cross entropy optimizer-adam,metrics-accuracy.

6. Model Training and Validation

- Training: Feed training data to the model, allowing it to learn patterns and relationships within cattle muzzle images.
- Validation: Assess the model's performance on the validation set to monitor for overfitting and fine-tune hyperparameters if needed.

7. Preprocessing for Prediction:

A function is defined to preprocess new images for prediction.
The image is loaded, converted to an array, expanded, and normalized.

8. Prediction : Finally the model takes the preprocessed input and predicts which cattle it belongs to.

Progress in project so far

We were able to procure a dataset and preprocess it.

Also we applied Data Augmentation and divided the dataset into train set and test set

A VGG16 model is loaded and used to extract Features(Beads and Ridges).

The model architecture involves a Convolution layer , MaxPool layer and fully connected layer (total 21 but only 16 with weights). The output layer has 1000 channels (one for every class)with a softmax activation function.

The model is compiled with the Adam optimizer, categorical cross-entropy loss, and balanced accuracy metric.

The model is trained with epoch 5

The trained model is used to predict cattle on the test set,and its graph is shown comparing the testing and validation performance..

Finally, the user can input any image of the cattle for its recognition.

The input image is preprocessed with the same size of 224,224 and passed through the model, and the predicted cattle along with index is given as output.

Testing and Validation

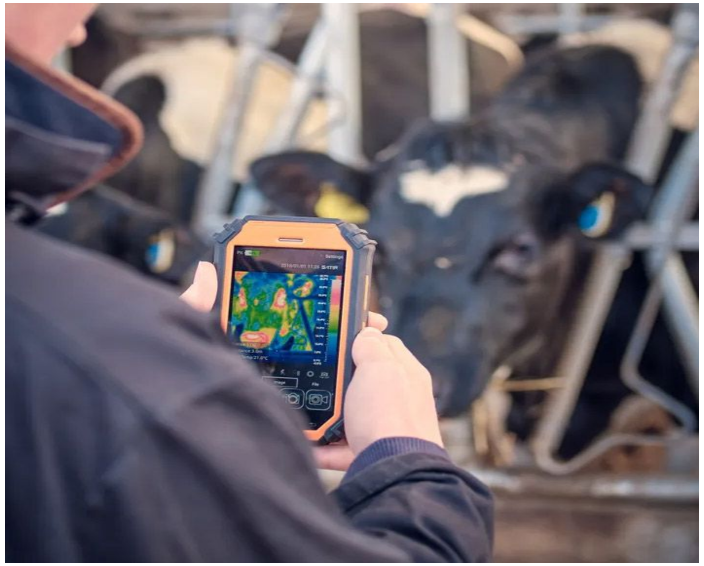
TEST CASE	GIVEN INPUT	EXPECTED OUTPUT	OBTAINED OUTPUT
Case Number 1	xyz1.jpg	Cattle-2	Index-1 Cattle-2
Case Number 2	xyz2.jpg	Cattle-7	Index-6 Cattle-7
Case Number 3	xyz3.jpg	Cattle-263	Index-262 Cattle-263
Case Number 4	xyz4.jpg	Cattle-67	Index-66 Cattle-67

Conclusion

"Muzzle pattern based cattle identification" using VGG16 architecture is making cattle identification easier and kinder to animals. We've created a smart system that can differentiate one cow from another by looking at their muzzle patterns. This means farmers don't have to do the hard work of tagging or branding cows anymore.

By using deep learning and the VGG16 architecture, we have developed a robust and accurate system that can uniquely identify cattle based on their muzzle patterns. We all care about the cows' feelings, so by using this technology we take a step forward to make their lives less stressful. This model is a big step forward in how we take care of cows and how farmers manage their herds.

As we keep working on this model, we're getting closer to a future where farming is both more accurate and more compassionate. This project shows how technology can change old farming methods for the better, helping both the animals and the farmers.



Reference

- Paper1: https://www.sciencedirect.com/science/article/abs/pii/S157495412100340X
- Paper2: https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/iet-ipr.2016.0799
- Paper3: https://idr-lib.iitbhu.ac.in/xmlui/bitstream/handle/123456789/2372/10_Chapter%2006.pdf?sequence=10&isAllowed=y
- Paper4: https://www.worldscientific.com/doi/10.1142/S0218001420560078
- https://www.kaggle.com/
- https://www.geeksforgeeks.org/
- https://www.w3schools.com/

Suggestions / Questions

Please …

Thank you !